

Smart Fish Control Command Guide

The smart fish features three operation modes: Basic Mode, Center Calibration Mode, and Custom Mode.

Upon power-on, the smart fish automatically enters Basic Mode.

1. Basic Mode (Default Mode)

Basic Mode is the default swimming mode, designed for direct control of the fish's movement.

Sending `CMD_D` or `CMD_E` to the fish allows entry into the other two debugging modes.

- **Function:** Execute preset actions (center return, left/right rotation, fixed-amplitude swing);
- **Entry Method:** Automatically entered after system power-on, or return by sending `X` in other modes.

Command	Function Description
<code>CMD_S</code>	Servo returns to the center position
<code>CMD_R</code>	Servo swings continuously to the right
<code>CMD_L</code>	Servo swings continuously to the left
<code>CMD_a</code>	±15° swing (low speed)
<code>CMD_A</code>	±15° swing (high speed)
<code>CMD_b</code>	±30° swing (low speed)
<code>CMD_B</code>	±30° swing (high speed)
<code>CMD_c</code>	±45° swing (low speed)
<code>CMD_C</code>	±45° swing (high speed)

This mode is used to calibrate the servo's center position.

Although the servo's center position is pre-calibrated to 90 degrees, deviations may occur over time. This mode is therefore used to determine the exact servo angle when the fish's tail is at the center position.

2. Center Calibration Mode

- **Function:** Customize the servo's center position (the default center can be calibrated via this mode);
- **Entry Method:** Send `CMD_D` in Basic Mode.

Operation	Function Description	Restrictions
Input a number (<code>CMD_0</code> - <code>CMD_180</code>)	Set the current servo angle as the new center	The angle must be within the range of 0-180 degrees; an error prompt will be issued if the range is exceeded
Send <code>CMD_X</code>	Return to Basic Mode	Residual input will be cleared before exiting to avoid interference

Note: The set angle will be stored in the flash memory of the main control chip, and the center angle will remain as the last set value upon next power-on.

Once the fish's center angle is determined, the swing can be controlled based on this position.
In this mode, you can set the servo to swing continuously between two specified angles. "Delay" can be understood as the number of milliseconds of delay per degree of servo movement. A larger value results in slower tail swing speed.

3. Custom Mode

- **Function:** Freely set the angle range of servo swing and the per-step delay time;
- **Entry Method:** Send `CMD_E` in Basic Mode.

Input Format 1

Note that three parameters separated by commas must be sent: `CMD_Angle1,Angle2,Delay(ms)`

- **Angle1/Angle2:** The two endpoint angles of the servo swing (e.g., 70 and 110; the servo will swing back and forth between 70° and 110°);
- **Delay(ms):** The per-step interval time for the servo to move from one angle to another (affects swing speed).

Parameter Restrictions

- Angle: 0-360 degrees (an error prompt will be issued if the range is exceeded);
- Delay: 1-1000ms (an excessively small value may cause servo jamming, while an excessively large value will reduce response speed).

Operation Commands

Operation	Function Description
Send parameters (e.g., <code>70,110,20</code>)	Swing cyclically according to the set angles and delay until a new command is sent
Send <code>x</code>	Return to Basic Mode and stop the current swing

Example:

Sending `CMD_70,110,20` → The serial port displays "✅ Custom swing: 70° ↔ 110°, delay 20ms", and the servo starts swinging back and forth;

Sending `CMD_0,180,500` → The servo swings slowly between 0° and 180°, with a per-step interval of 500ms.

Input Format 2

Comparison with real fish: A real fish's tail does not swing continuously; after a few swings, the fish pauses to rest.

Custom Mode supports simulating the fish's pause time.

Note that five parameters separated by commas must be sent:

`CMD_Angle1,Angle2,Delay(ms),Swing Cycles,Pause Duration`

- **Angle1/Angle2:** The two endpoint angles of the servo swing (e.g., 70 and 110; the servo will swing back and forth between 70° and 110°);
- **Delay(ms):** The per-step interval time for the servo to move from one angle to another (affects swing speed);
- **Swing Cycles:** The number of cycles the servo swings;
- **Pause Duration:** The pause time (in ms) after completing N cycles (where N is the previous parameter).

Operation Commands

Operation	Function Description
Send parameters (e.g., <code>CMD_70,110,20,3,2000</code>)	Swing between 70° and 110° with a per-step delay of 20ms (controlling speed); after 3 swing cycles, pause for 2 seconds, then resume swinging for 3 cycles, and repeat
Send <code>CMD_X</code>	Return to Basic Mode and stop the current swing